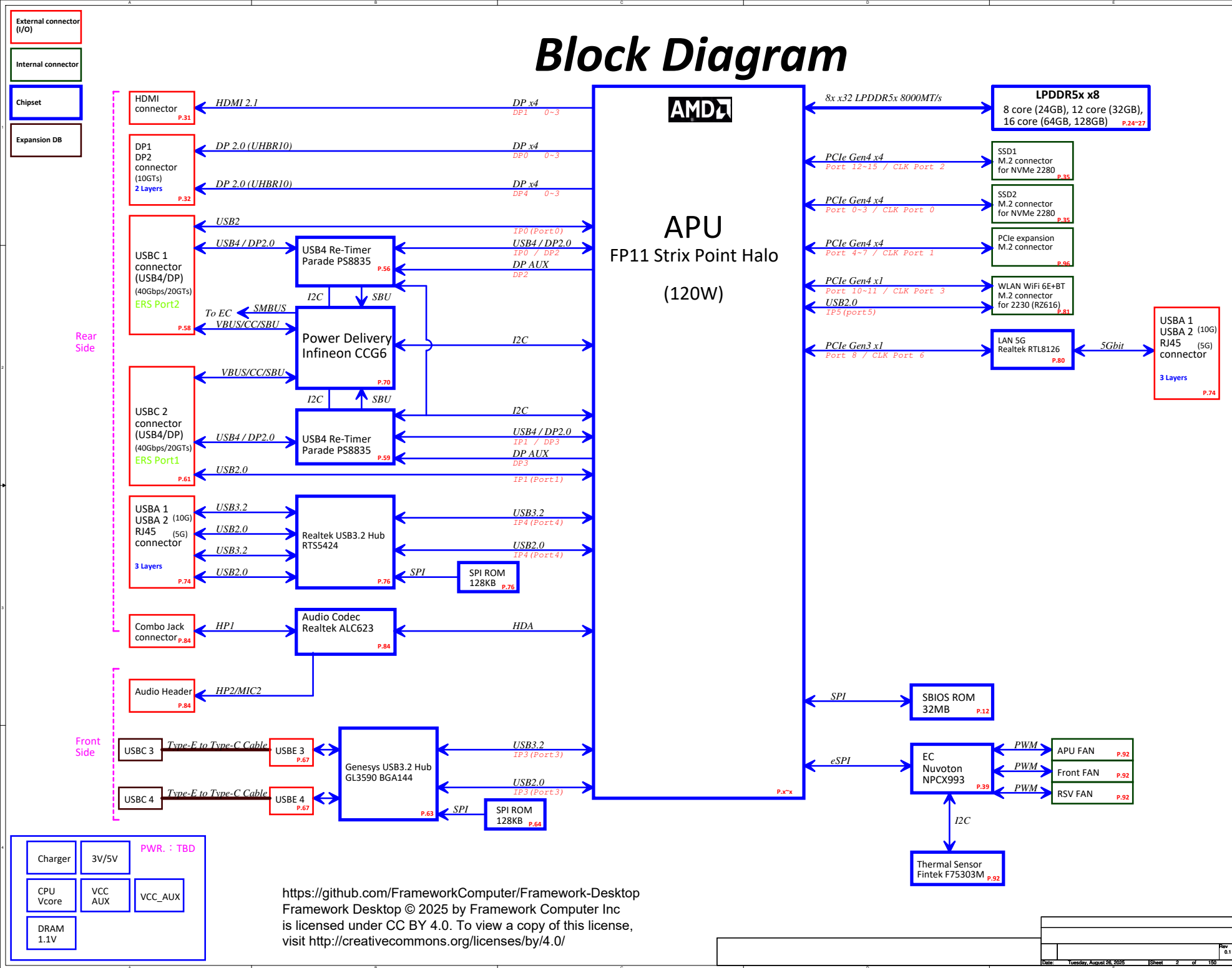
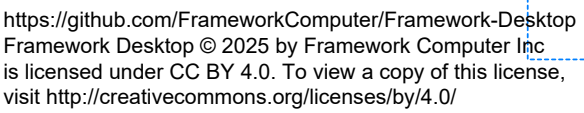
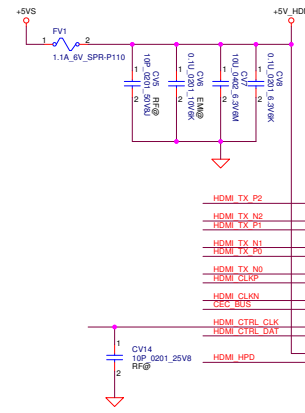
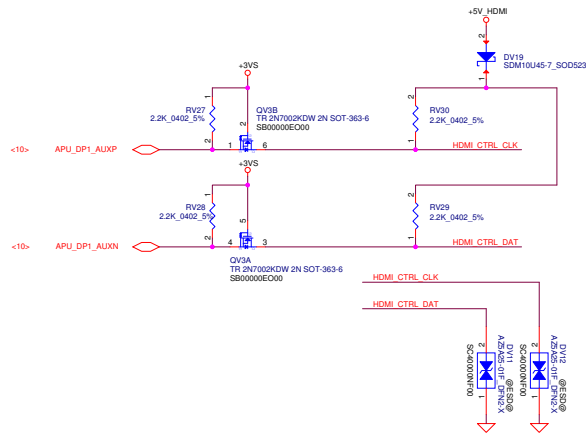
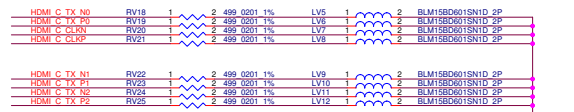
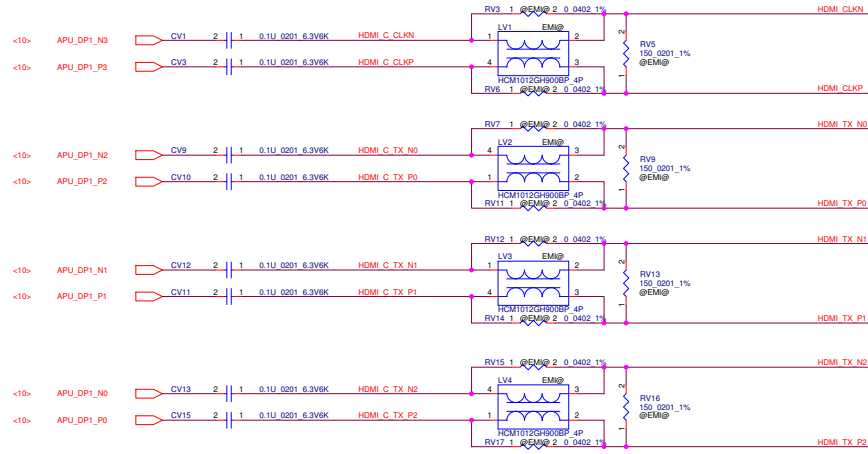


Block Diagram

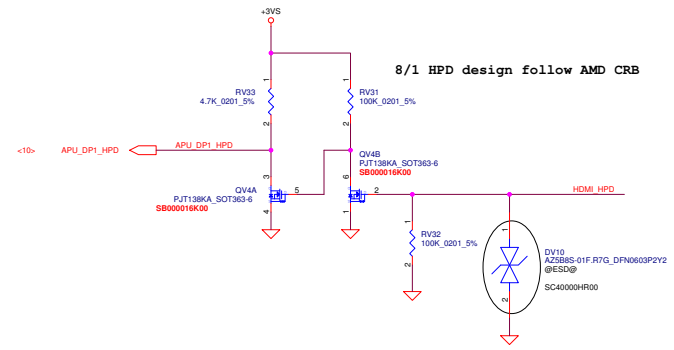
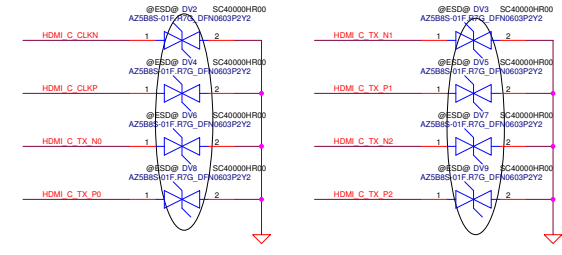
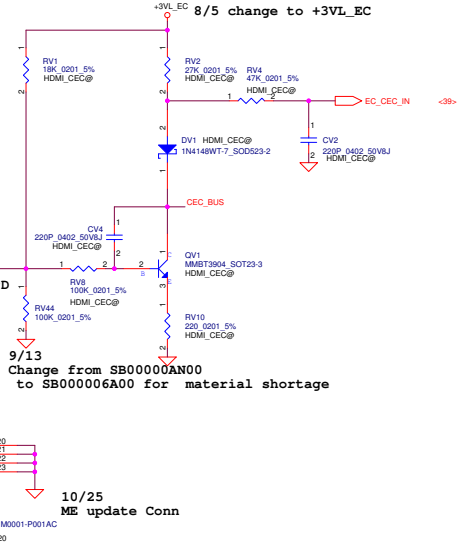




04/08 PVT, RV3, RV6, RV7, RV11, RV12, RV14, RV15, RV17, Solder Mask



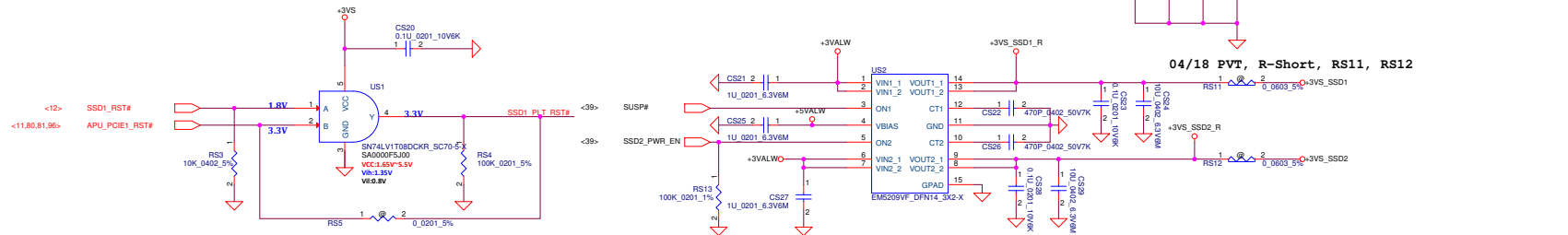
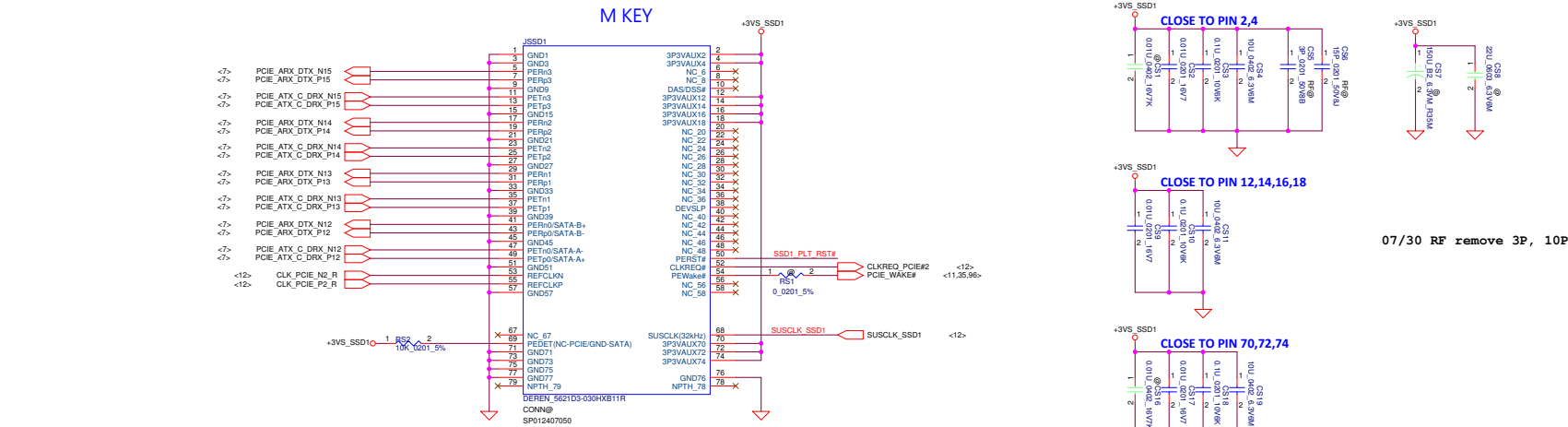
HDMI_CEC



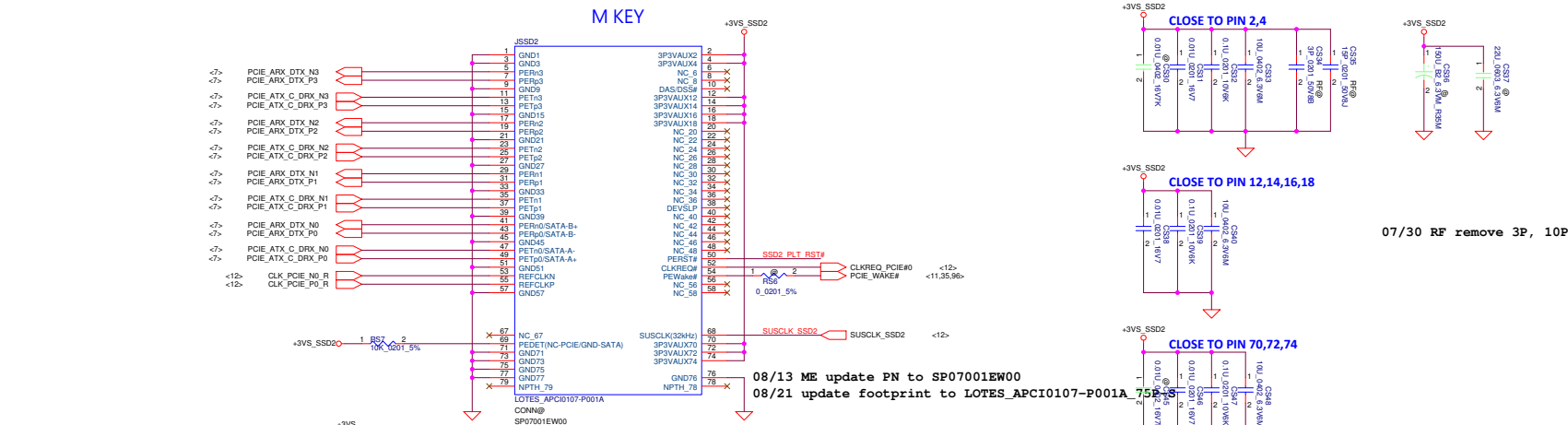
8/1 HPD design follow AMD CRB

Compal Electronics, Inc.	
Title	P031-HDMI_EE
Document Number	LA-0841P
Date	Tuesday, August 28, 2024
Sheet	31 of 150

M KEY



M KEY



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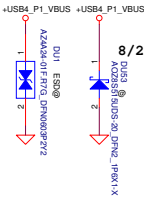
Compal Electronics, Inc.		Title	
P035-NGFF SSD x2 EE		Document Number	
LA-0841P		Rev	
Date: Tuesday, August 26, 2025		Sheet 35 of 150	

04/08 PVT, RU51, RU52, Solder Mask

08/09 EMI change to mount

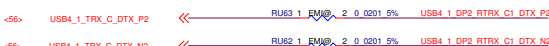
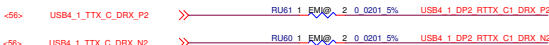
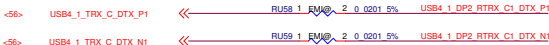
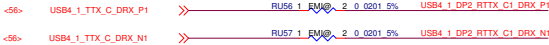
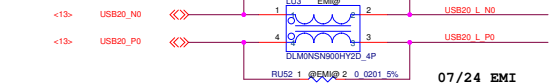
08/07 Layout Swap

07/24 EMI



8/2 Follow Lilac

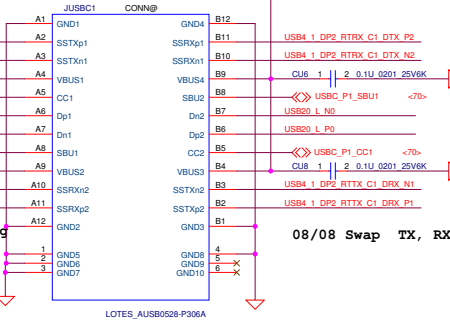
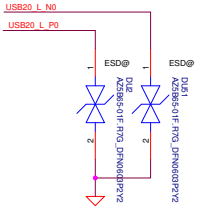
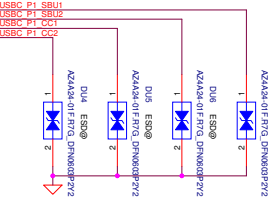
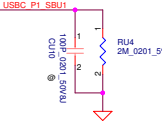
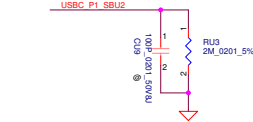
07/29 follow Galio AMD



08/06 Remove choke for follow PS8835 CRB

08/18 Change CU5, CU6, CU7, CU8 from 0603_100V to 0201_25V for no charging

08/14 Pin5, PIN6 is NPTH

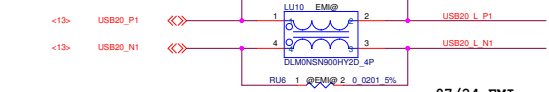


08/08 Swap TX, RX, CC, SBU 1 / 2

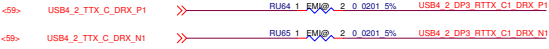
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04/08 PVT, RU5, RU6, Solder Mask

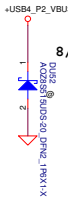
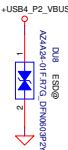
08/09 EMI change to mount



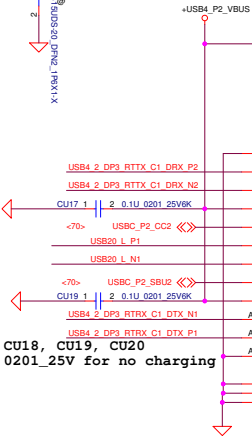
07/24 EMI



08/06 Vendor ask to remove choke



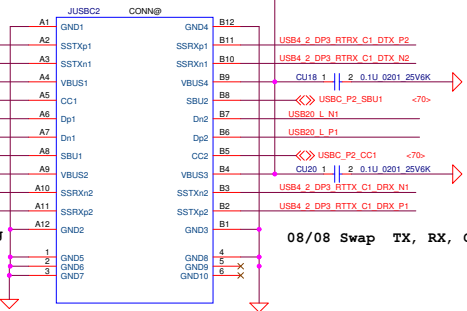
8/2 Follow Lilac



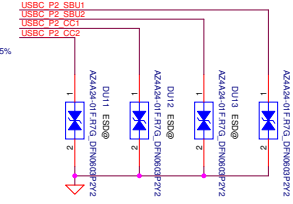
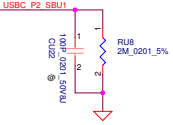
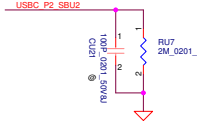
08/18 Change CU17, CU18, CU19, CU20 from 0603_100V to 0201_25V for no charging

08/14 Pin5, PIN6 is NPTH

07/29 follow Galio AMD



08/08 Swap TX, RX, CC, SBU 1 / 2



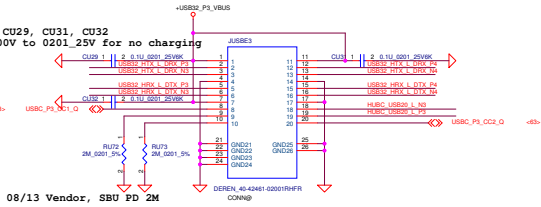
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Main Function: JUSBC1/2

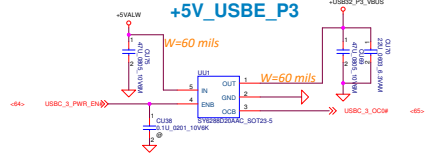
04/08 PVT, RU9, RU10, RU11, RU13, RU14, RU15, RU16, RU17, RU20, Solder Mask



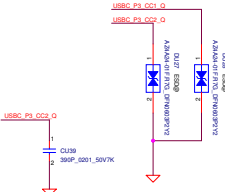
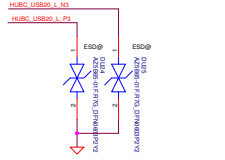
08/18 Change CU29, CU31, CU32
from 0603_100V to 0201_25V for no charging



08/13 Vendor, SBU PD 2M

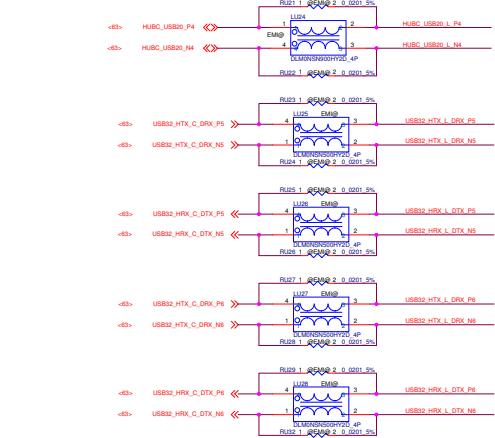


07/27 ESD remove DU16 ~ DU23

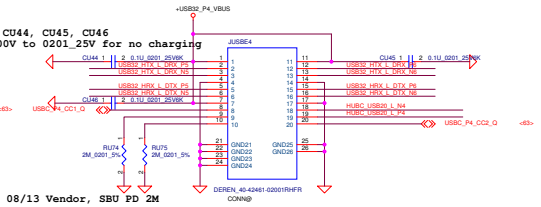


Main Function: JUSBC1/2

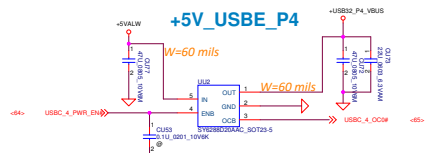
04/08 PVT, RU21, RU22, RU23, RU24, RU25, RU26, RU27, RU28, RU29, RU32, Solder Mask



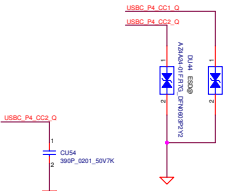
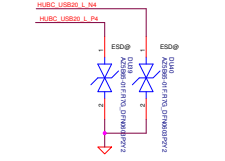
08/18 Change CU44, CU45, CU46
from 0603_100V to 0201_25V for no charging



08/13 Vendor, SBU PD 2M

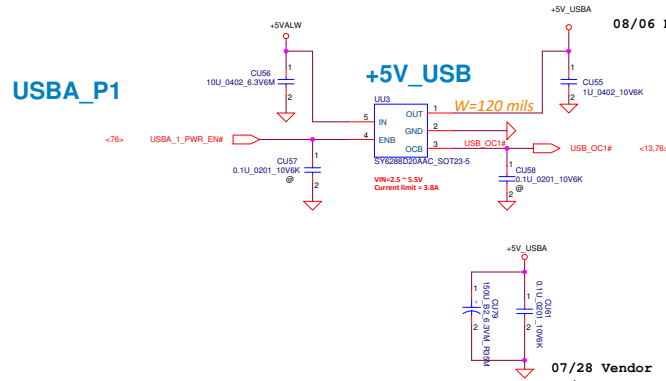


07/27 ESD remove DU31 ~ DU38

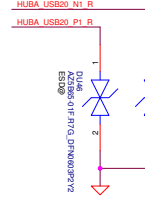
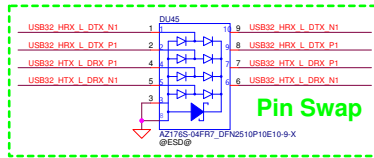
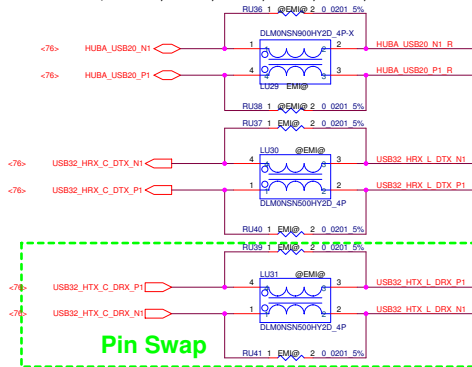


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Yr	P067-USBE P3P4 Conn EXP EE
Rev	1A-0841P
Rev	6.1

USBA_P1



04/08 PVT, RU36, RU38, LU30, LU31, Solder Mask



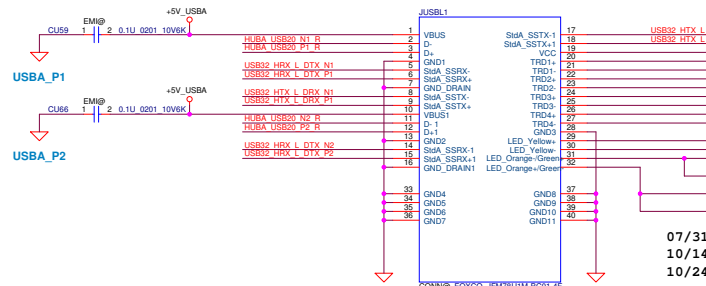
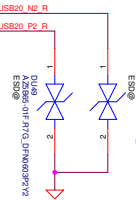
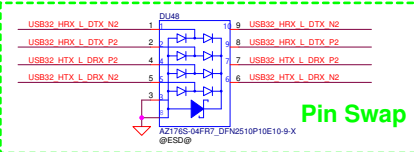
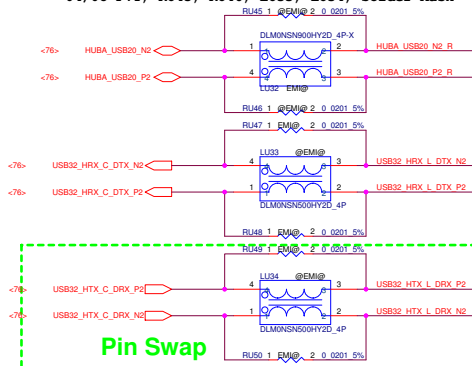
07/28 Vendor
10/30 Update CU79 from 100u to 150u

Singatron

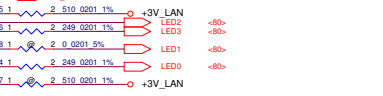


03/17 Remove LU30, LU31, LU33, LU34
Add RU37, RU40, RU39, RU41, RU47, RU48, RU49, RU50

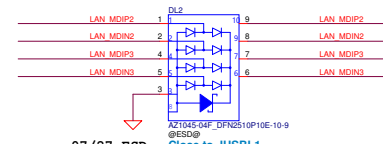
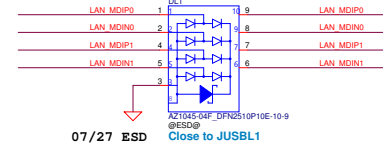
04/08 PVT, RU45, RU46, LU33, LU34, Solder Mask



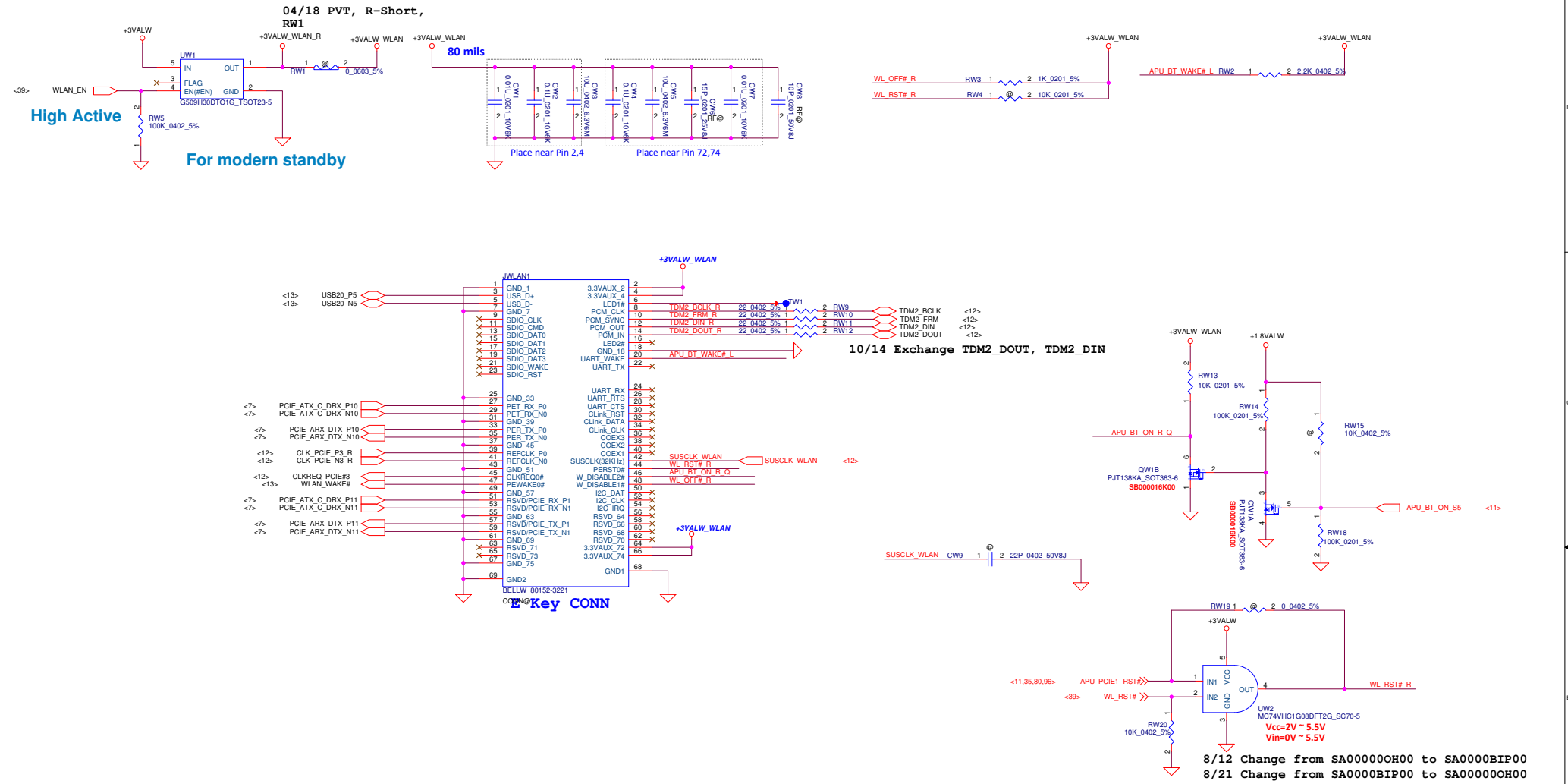
07/29 LED follow Singatron spec,
reverse Orange/Green



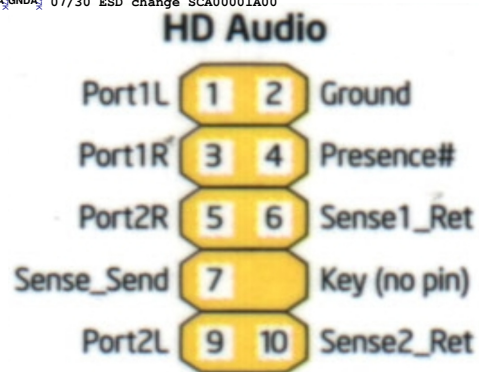
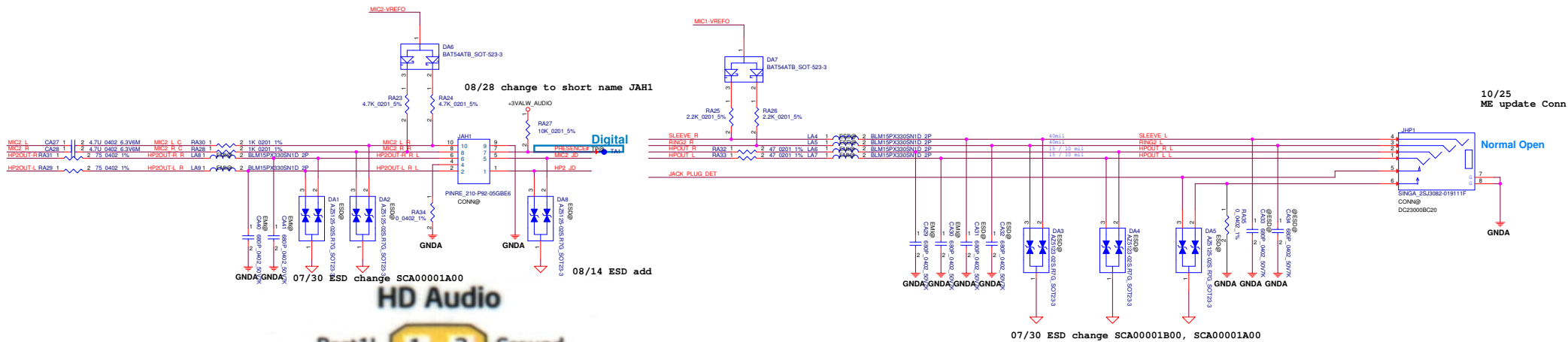
07/31 Confirm LED green light with Singatron RJ45
10/14 Change pin 32 to connect LED0
10/24 Change LED0 to LED3 for G/Y light switch, reserve EVT design +3V_LAN to LED1



Main Func = WLAN M.2



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		Compal Electronics, Inc.	
		Title	
		P084-Codex EE	
		Document Number	
		LA-0841P	
		Date	Tuesday, August 26, 2025
		Sheet	84 of 150
		Rev	0.1

Main Func = LED

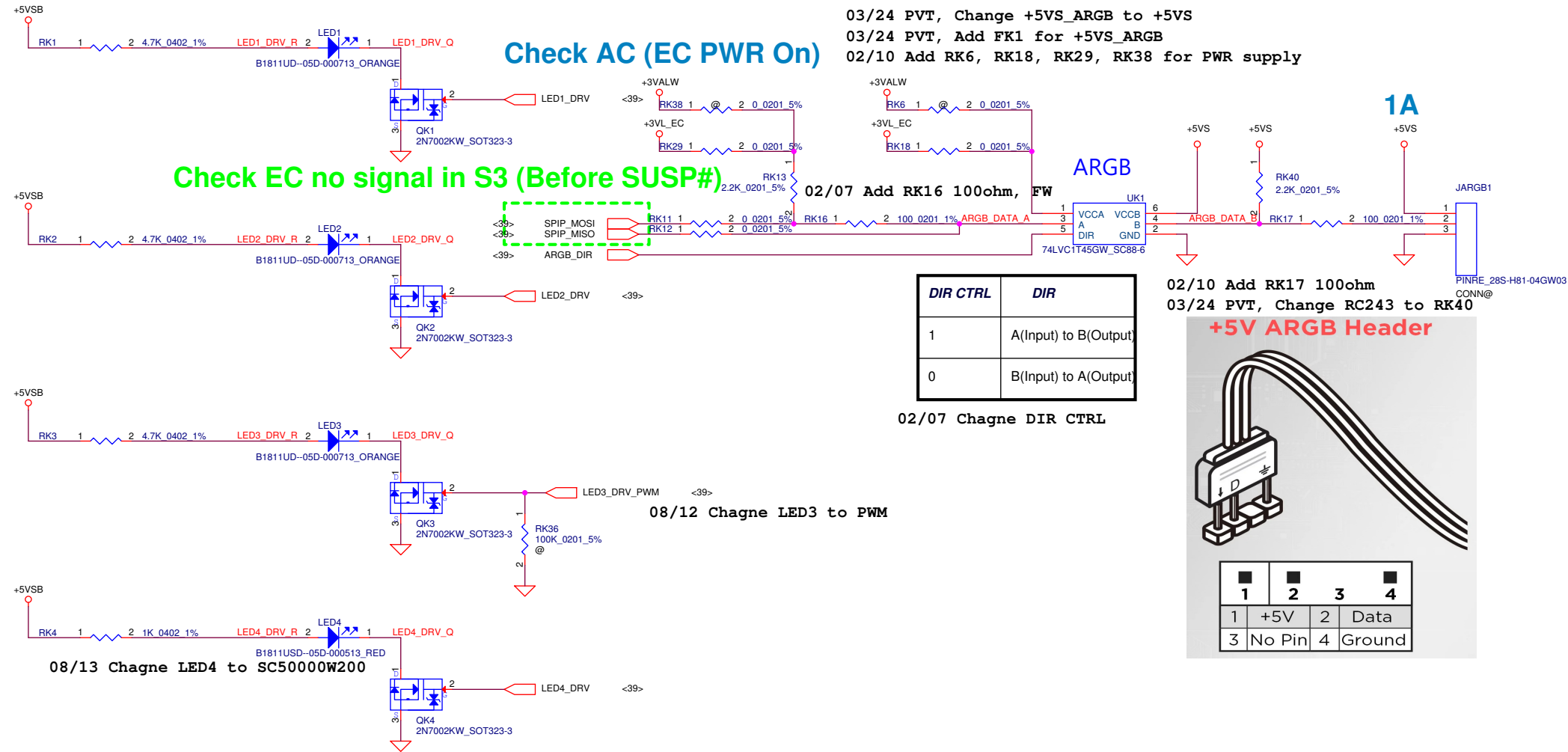
03/24 Need to update PN, SYMBOL, FK1, SP04000FV00, Change to FZ1

08/13 Chagne RK1, RK2, RK3, RK4 to 604ohm for current 5mA
PVT, 06/16 Chagne RK1, RK2, RK3, to 4.7K, SD034470180

Check AC (EC PWR On)

Check EC no signal in S3 (Before SUSP#)

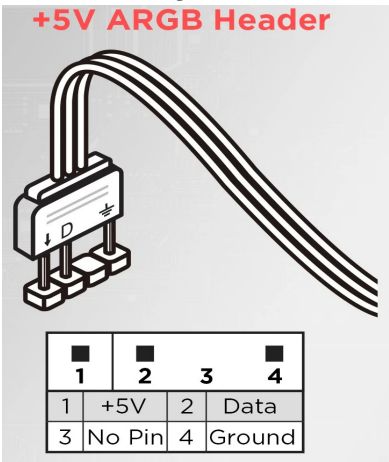
03/24 PVT, Change +5VS_ARGB to +5VS
03/24 PVT, Add FK1 for +5VS_ARGB
02/10 Add RK6, RK18, RK29, RK38 for PWR supply



DIR CTRL	DIR
1	A(Input) to B(Output)
0	B(Input) to A(Output)

02/07 Chagne DIR CTRL

02/10 Add RK17 100ohm
03/24 PVT, Change RC243 to RK40



08/13 Chagne LED4 to SC50000W200

08/12 Chagne LED3 to PWM

05/08 PVT, Chagne LED resistance RK1, RK2, RK3, RK4 to 1K

Compal Electronics, Inc.			
Title			
P090-LED/ARGB_EE			
Document Number			
LA-0841P			
Rev			
0.1			
Date:	Tuesday, August 26, 2025	Sheet	90 of 150

0

<39> PWM_FAN1

+3VS

10K/0.001.5% R_F3

1 2 3 4 5 6

QF2B
PJT138KA_SOT363-6
SB000016K00

+3VS

10K/0.001.5% R_F10

1 2 3 4 5 6

QF2A
PJT138KA_SOT363-6
SB000016K00

<39> PWM_FAN2

07/30 FM for prevent

2.4A

FF1

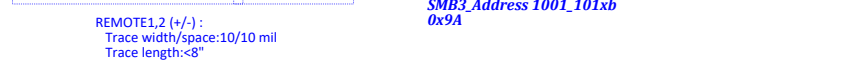
1 2

1.5A_12V_PBL0805-250-12

Vmax : 12V
Hold Current : 2.5A



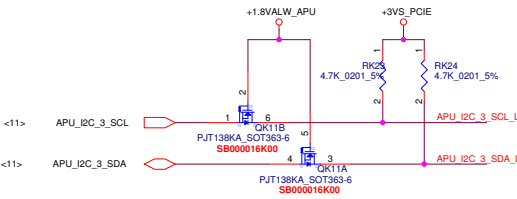
07/30 RF remove 6P, 8



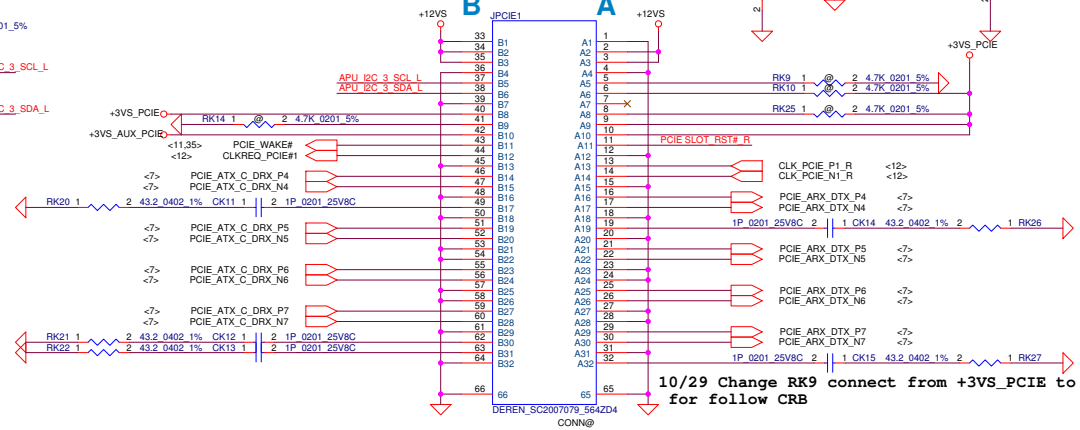
	2	1
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Main Function:



10/08 PCIe Conn Symbol Mirror Horizontally



8/12 Change from SA00000H00 to SA0000BIP00
8/21 Change from SA0000BIP00 to SA00000H00

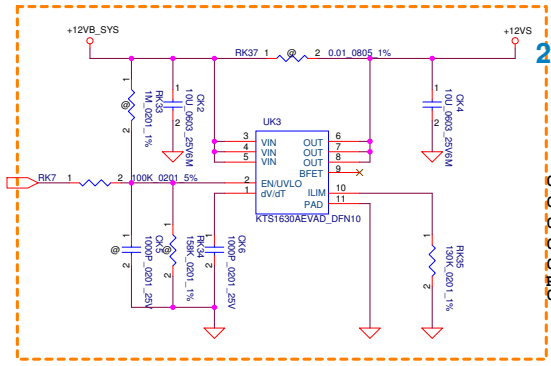
引脚	B侧	A侧	描述
1	+12 V	PRSNT1#	必須連接到系統PRSNT2# 引脚
2	+12 V	+12 V	
3	+12 V	+12 V	
4	Ground	Ground	
5	SMCLK	TCK	
6	SMDAT	TDI	
7	Ground	TDO	SMBus和JTAG 端引脚
8	+3.3 V	TMS	
9	TRST#	+3.3 V	
10	+3.3 V aux	+3.3 V	待机电源
11	WAKE#	PERST#	鏈結啟用; 基本復位
Key notch			
12	CLKREQ#	Ground	要求運行的時鐘
13	Ground	REFCLK+	參考時鐘差分對
14	HSOp (0)	REFCLK-	
15	HSOn (0)	Ground	
16	Ground	HSIp (0)	通道0接收數據; 十和-
17	PRSNT2#	HSIn (0)	
18	Ground	Ground	
PCI Express x1卡於引脚18結果			
19	HSOp (1)	Reserved	通道1傳輸數據; 十和-
20	HSOn (1)	Ground	
21	Ground	HSIp (1)	通道1接收數據; 十和-
22	Ground	HSIn (1)	
23	HSOp (2)	Ground	通道2傳輸數據; 十和-
24	HSOn (2)	Ground	
25	Ground	HSIp (2)	通道2接收數據; 十和-
26	Ground	HSIn (2)	
27	HSOp (3)	Ground	通道3傳輸數據; 十和-
28	HSOn (3)	Ground	
29	Ground	HSIp (3)	通道3接收數據; 十和-
30	Reserved	HSIn (3)	
31	PRSNT2#	Ground	
32	Ground	Reserved	

4.1. Power Supply Requirements

All PCI Express add-in card connectors require two power rails: +12V and +3.3V, with a third, optional 3.3Vaux rail. Systems that provide PCI Express add-in card connectors are required to provide both the +12V and +3.3V rails to every PCI Express add-in card connector in the system. The 3.3Vaux rail may be supplied to the PCI Express add-in card connectors at the system board designers' discretion. However, if a system board designer does supply 3.3Vaux to the PCI Express add-in card connectors, the 3.3Vaux rail must be supplied to all PCI Express add-in card connectors. In addition, as described in Chapter 2, if the platform with the PCI Express interface supports the WAKE# signal, the 3.3Vaux rail (as well as the WAKE# signal) must be supplied to all PCI Express add-in card connectors.

Table 4-1 provides the required specifications for the power supply rails available at the PCI Express connectors, based on the number of PCI Express lanes supported by the connectors. The system designer is responsible for ensuring that the power delivered to the PCI Express connectors meets the specifications called out in Table 4-1.

Table 4-1: Power Supply Rail Requirements			
Power Rail	x1 Connector	x4/x8 Connector	x16 Connector
+3.3V			
Voltage tolerance	± 9% (max)	± 9% (max)	± 9% (max)
Supply Current	3.0 A (max)	3.0 A (max)	3.0 A (max)
Capacitive Load	1000 µF (max)	1000 µF (max)	1000 µF (max)
+12V			
Voltage tolerance	± 8% (max)	± 8% (max)	± 8% (max)
Supply Current	0.5 A	2.1 A (max)	4.4 A (max)
Capacitive Load	300 µF (max)	1000 µF (max)	2000 µF (max)

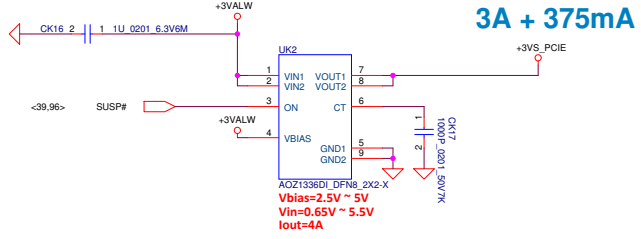


2.1A + 2.4A

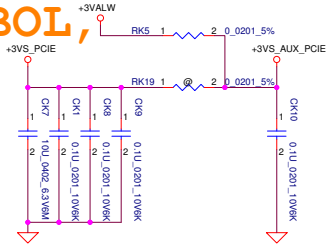
08/15 Follow CRB 0.5K
09/02 Change to SD034499080 for material shortage
03/24 PVT, Change from PMOS to G3720, SA0000IQY00
03/31 PVT, Add RK7, Change CK5, CK6 to SE170102K80
04/8 PVT, nun-mon RK33, RK34, RK35 change to 130K
04/9 PVT, change CK2, CK4 to 10u for KTS1630 design

01/16 Add RK5 for PCIE_AUX connect to S5 PWR, DVT2

03/24 Need to update PN, SYMBOL, UK3, SA0000IRO00



3A + 375mA



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Input and Output current from Power budget

Vin:+12VB_APU
Input current:29.802A

Vin:+12VB_SYS
Input current:4.6A

Vin:+3VALW
Input current:9.59A

Vin:+5VALW
Input current:10.53A

PUS_CON
8-Pin
24-Pin

+12VB_APU_R

+12VB_APU

+12VB_SYS_R

+12VB_SYS

+5V_PSU

+3V_PSU

+3VIN_PSU

+5VIN_PSU

+5VSB

+12VB_SYS_R

+12VB_SYS

+5VSB

+12VB_SYS

+5VSB

+12VB_SYS

+5VSB

+12VB_SYS

+5VSB

+12VB_SYS

+5VSB

+12VB_SYS

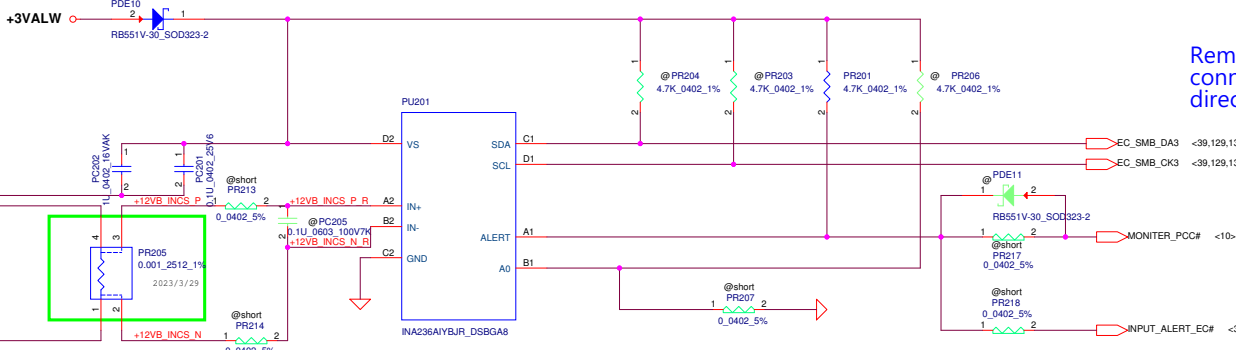
+5VSB

+12VB_SYS

+5VSB

+12VB_SYS

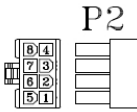
+5VSB



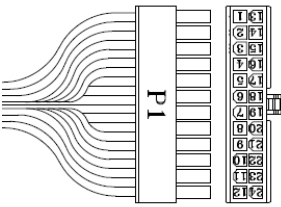
Remove RG63/QG5 and connect OC_WARN# directly by OD.

Table 7-1. Address Pins and Target Addresses

Address	Target Address
GND	1000000
VS	1000001
SDA	1000010
SCL	1000011



Pin	Signal	Color	IO
1	COM	BLACK	18
2	COM	BLACK	18
3	COM	BLACK	18
4	COM	BLACK	18
5	+12V	YELLOW	18
6	+12V	YELLOW	18
7	+12V	YELLOW	18
8	+12V	YELLOW	18

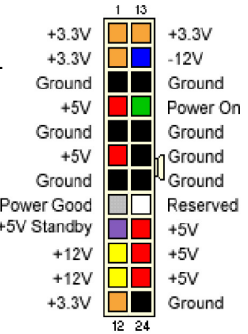
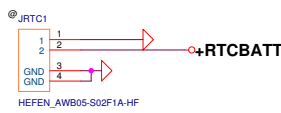


Pin	Signal	Color	IO
1	+3.3V	ORANGE	18
2	+3.3V	ORANGE	18
3	COM	BLACK	18
4	+5V	RED	18
5	COM	BLACK	18
6	+5V	RED	18
7	COM	BLACK	18
8	PW-OK	GRAY	24
9	PW-OK	GRAY	24
10	+5Vsb	PURPLE	24
11	+5Vsb	PURPLE	24
12	+12V	YELLOW	18
13	+3.3V	ORANGE	18
14	+3.3V	ORANGE	18
15	+3.3V	ORANGE	18
16	+3.3V	ORANGE	18
17	+3.3V	ORANGE	18
18	+3.3V	ORANGE	18
19	+3.3V	ORANGE	18
20	+3.3V	ORANGE	18
21	+3.3V	ORANGE	18
22	+3.3V	ORANGE	18
23	+3.3V	ORANGE	18
24	+3.3V	ORANGE	18

4.1.1 OUTPUT RATING

Output	Nominal	Regulation	Ripple/Noise	Min	Max
1	+3.3V	±5%	50 mV	0A	14.0 A
2	+5V	±5%	50 mV	0A	16.0 A
3	+12V	±5%	120 mV	0.05A	36.0 A
4	-12V	±10%	120 mV	0 A	0.3A
5	+5Vsb	±5%	50 mV	0 A	3A

* ±12V Peak load 600W/10us (Capacitive Load 4700 µF)



ATX 24-pin power connector

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